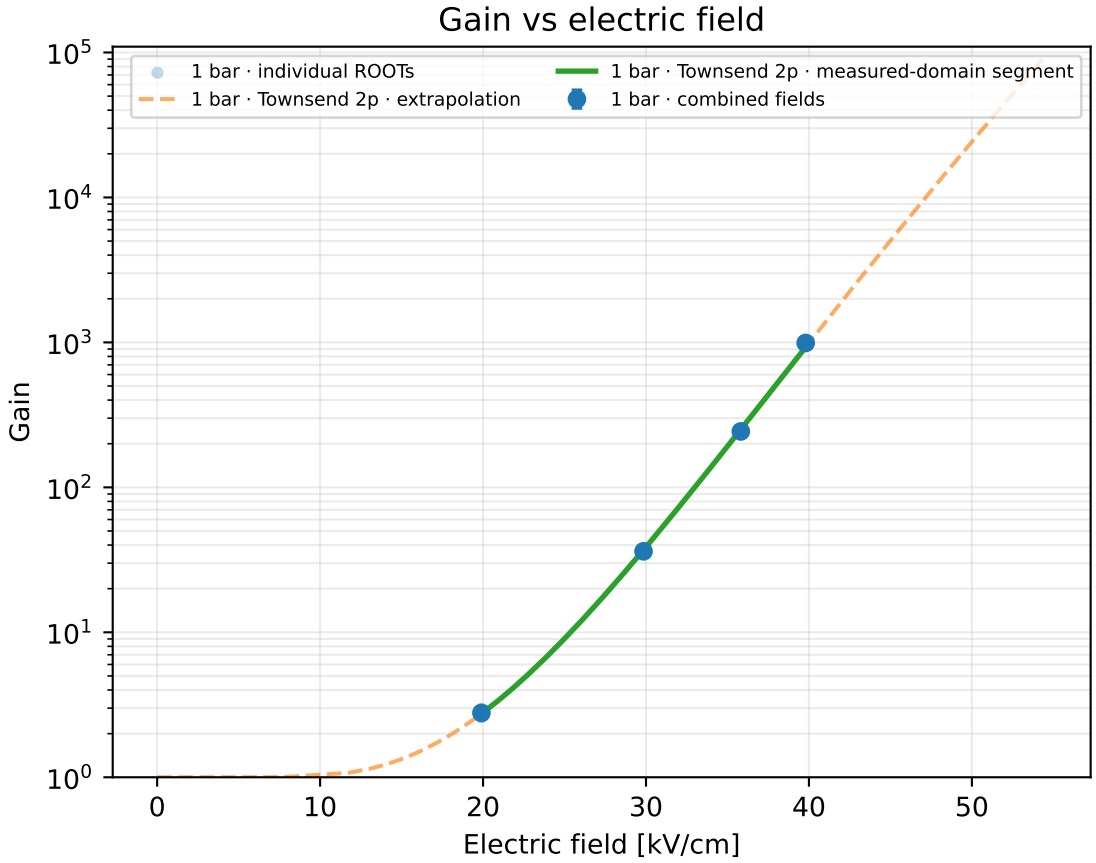
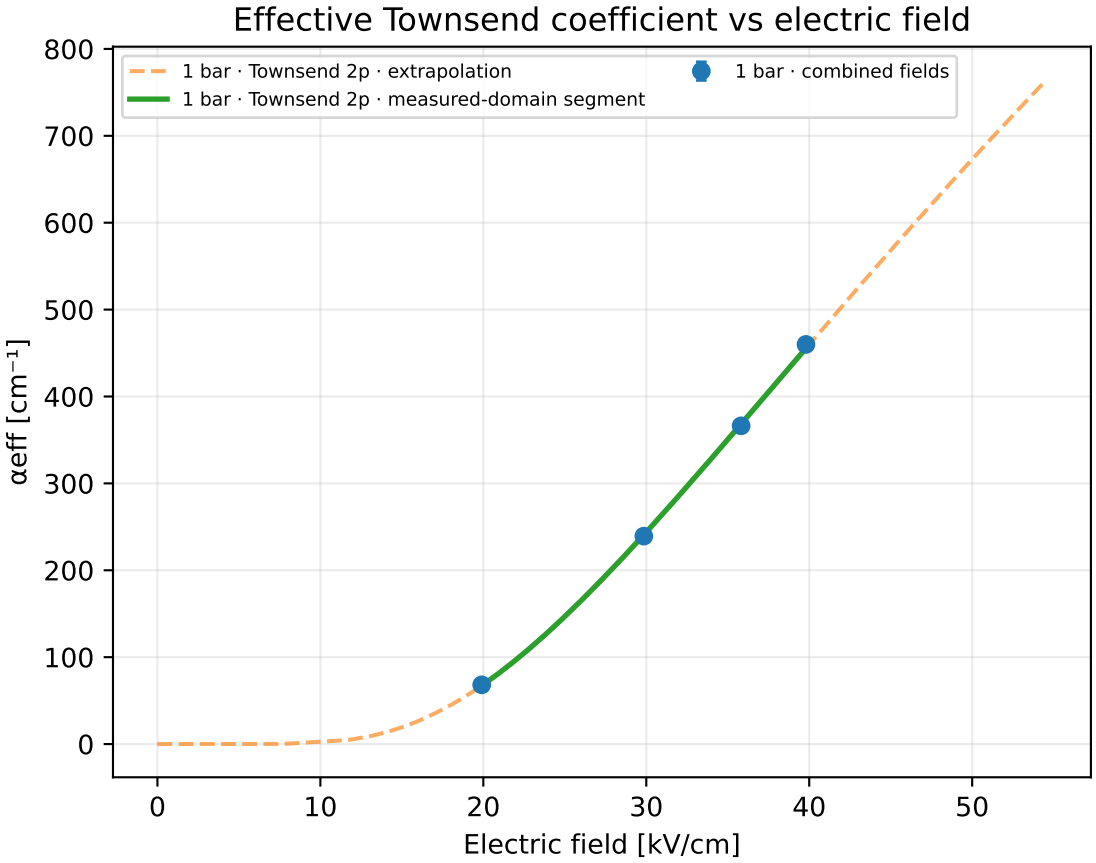
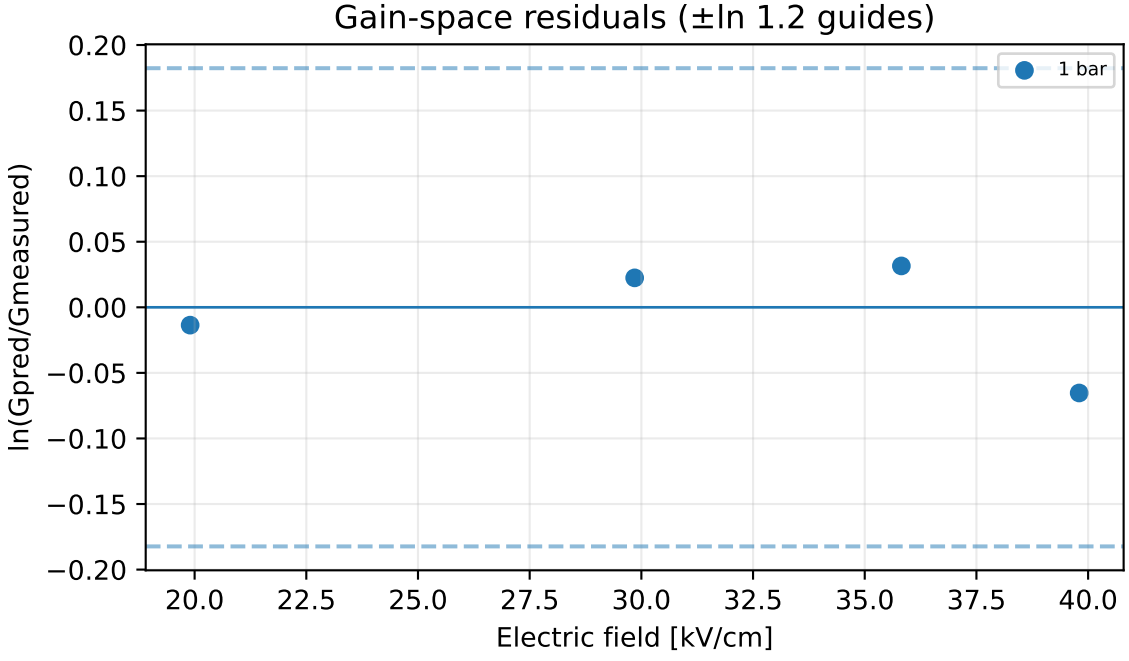
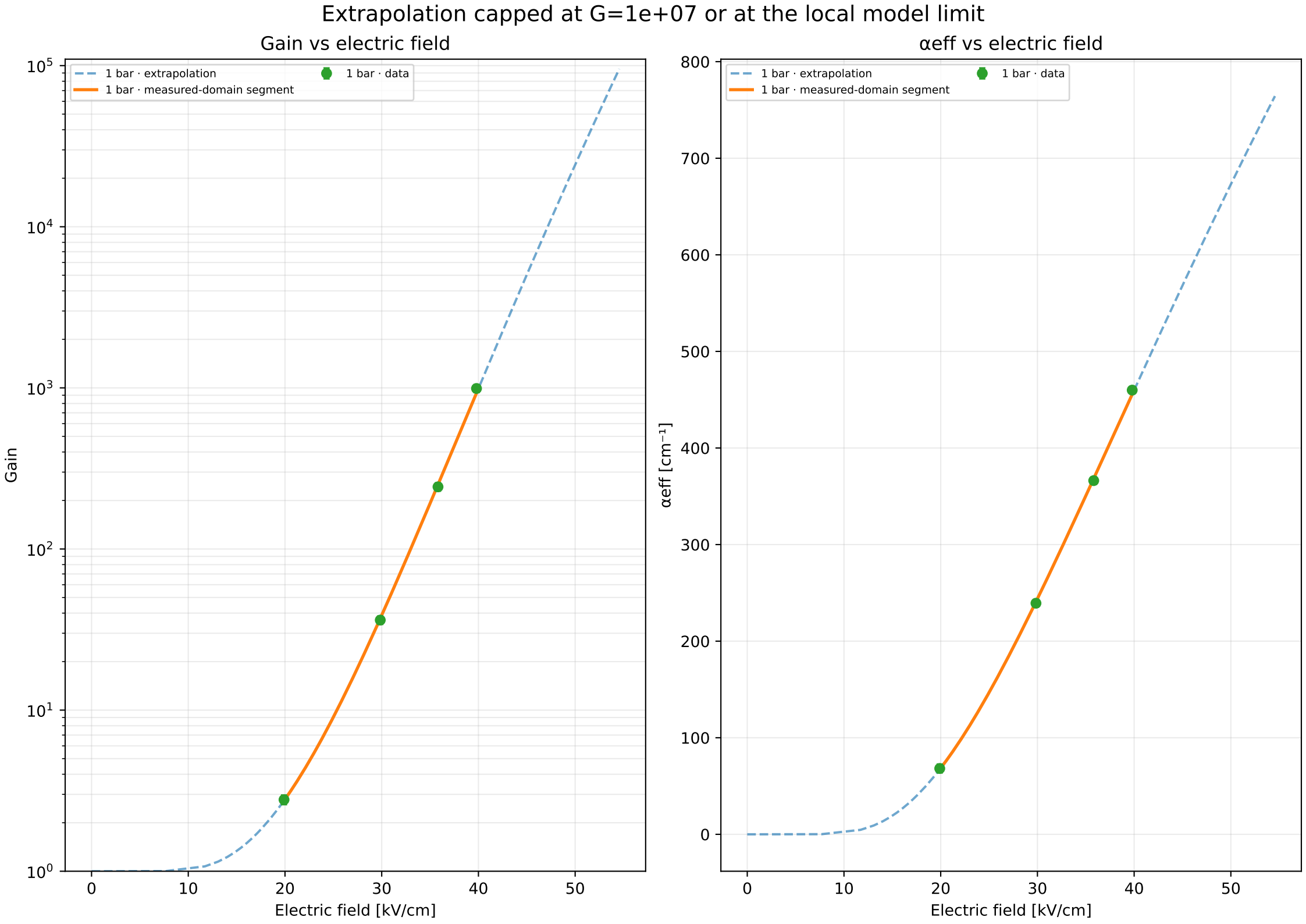
ArCF4 · Ar 90 % / CF_4 10 % · gap 0.15 mm








```
ROOT runs: 8
Unique physical points: 4
Pressures: 1 bar
Selected model: Townsend 2p
Status: rejected_poor_cross_validation
Training RMSE: ×1.009
Cross-validation median: unavailable
Cross-validation worst: unavailable
Reduced-field range: 19.9–39.8 kV cm⁻¹ bar⁻¹
A=3085.8, B=76.138, m=0, n=1
Covariance condition: 189
Warnings: poor_cross_validation, large_cross_validation_outlier
1 bar display limit: G=9.54e+04 (local inversion limit)
```


Extrapolation capped at G=1e+07 or at the local model limit
Gain vs electric field
1 bar · extrapolation
1 bar · measured-domain segment
1 bar · data
Gain
Electric field [kV/cm]
αeff vs electric field
αeff [cm⁻¹]
Electric field [kV/cm]

Exact combined points used by the fit · page 1/1

Repeated ROOTs at the same field are pooled once; all source ROOTs remain on disk.

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm⁻¹]	npe	runs
1	19.9	19.9	2.78	0.189	68.1634	200	2
1	29.85	29.85	36.205	2.14	239.28	200	2
1	35.82	35.82	243.09	14.5	366.229	200	2
1	39.8	39.8	991.125	56.4	459.923	200	2

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-21.5	—	—	no	poor cross validation, large cross validation outlier